

Lab Assignment 05



Inspiring Excellence



Course Code:	CSE111
Course Title:	Programming Language II
Topic:	Access Modifier & Encapsulation
Number of Tasks:	10 (Classwork: 04, Homework: 06)

[Submit all the Coding Tasks (Homework: Task 1 to 4) in the Google Form shared on buX before the next lab. Submit the Tracing Tasks (Homework: Task 5 to 6) handwritten to your Lab Instructors at the beginning of the lab]

[You are not allowed to change the driver codes of any of the tasks]

CLASSWORK

Task 1

Carefully observe the following driver code and expected output. Write the java code for the **MFS** Class and **Card** Class both. Consider that there can be at most 1 card per MFS account.

[balance attribute of MFS class will be **Private**]

Tester Class	Expected Output
<pre>public class CardTester{ public static void main(String [] args){ System.out.println("1-----"); MFS bkash = new MFS("12121212"); System.out.println("2-----"); Card visaCard = new Card(2000, "VISA"); System.out.println("3-----"); bkash.addMoneyFromCard(5000); System.out.println("4-----"); visaCard.addMoneyFromAccount(500, bkash); System.out.println("5-----"); bkash.addNewCard(visaCard); System.out.println("6-----"); bkash.addMoneyFromCard(5000); System.out.println("7-----"); bkash.addMoneyFromCard(800); System.out.println("8-----"); visaCard.addMoneyFromAccount(500, bkash); } }</pre>	<pre>1----- Account created. AccNo: 12121212 Balance: 0.0 tk Card not saved. 2----- VISA Card created successfully. Balance: 2000.0 tk 3----- Please add a card to your MFS account 4----- Insufficient account balance! 5----- VISA Card added to MFS acc#12121212 6----- Insufficient balance in VISA Card. 7----- Transfer successful. New MFS account balance: 800.0 tk. Card balance: 1200.0 tk 8----- Added 500 tk to card from MFS account#12121212. New balance in card: 1700.0 tk</pre>

Task 2

Design the **necessary** classes for the NetFlix system with the necessary properties to produce the given output for the provided driver code.

[All attribute of Movie class will be **Private**]

Driver code	Output
<pre>public class NetflixAccount{ public static void main(String[] args){ Movie m1 = new Movie("Avengers", "Sci-fi", "1:59:08"); Movie m2 = new Movie("Lord of the Rings", "Fantasy", "2:58:05"); Movie m3 = new Movie("Final Destination","Horror"); Movie m4 = new Movie("Fast and Furious","Action", "2:05:04"); System.out.println("====="); NetflixUser user1 = new NetflixUser("Martin Freeman", "Free"); user1.addToFavourites(m1); user1.showFavourites(); user1.upgradePlan(); user1.addToFavourites(m1); user1.showFavourites(); System.out.println("====="); NetflixUser user2 = new NetflixUser("Vin Diesel", "Premium"); user2.addToFavourites(m1); user2.addToFavourites(m2); user2.addToFavourites(m3); user2.addToFavourites(m4); user2.showFavourites(); System.out.println("====="); NetflixUser user3 = new NetflixUser("Chris Evans"); user3.addToFavourites(m1); user3.upgradePlan(); user3.addToFavourites(m2); user3.addToFavourites(m3); user1.upgradePlan(); } }</pre>	<pre>===== New user account created! Cannot add movies to favourites. Please upgrade to Netflix Premium. No available favourites. Please upgrade to Netflix Premium. Welcome to Netflix Premium! Movie added to favourites. Showing Martin Freeman's Favourites: Avengers, Sci-fi, Duration: 1:59:08 ===== New user account created! Movie added to favourites. Movie added to favourites. Movie added to favourites. Cannot add new movies. Favourites list is full. Showing Vin Diesel's Favourites: Avengers, Sci-fi, Duration: 1:59:08 Lord of the Rings, Fantasy, Duration: 2:58:05 Final Destination, Horror, Duration: Unknown ===== New user account created! Cannot add movies to favourites. Please upgrade to Netflix Premium. Welcome to Netflix Premium! Movie added to favourites. Movie added to favourites. You already have Netflix Premium!</pre>

Task 3

Design the required class/es so that the following output is generated. Read the following description:

1. You may assume that to board a bus, a student must have the bus pass, and his/her location must match the route of the bus.
2. Additionally, the default maximum capacity of the bus is 2.
3. The location attribute of the Student class will be **private**

Driver Code	Output
<pre>public class BracuStudentTester { public static void main(String[] args) { BracuStudent st1 = new BracuStudent("Afif", "Mirpur"); System.out.println("1====="); BracuStudent st2 = new BracuStudent("Shanto", "Motijheel"); BracuStudent st3 = new BracuStudent("Taskin", "Mirpur"); st1.showDetails(); st2.showDetails(); System.out.println("2====="); st3.showDetails(); System.out.println("3====="); BracuBus bus1 = new BracuBus("Mirpur"); BracuBus bus2 = new BracuBus("Azimpur", 5); bus1.showDetails(); bus2.showDetails(); System.out.println("4====="); st2.collectPass(); st3.collectPass(); System.out.println("5====="); st2.showDetails(); st3.showDetails(); System.out.println("6====="); bus1.board(); System.out.println("7====="); bus1.board(st1, st2); System.out.println("8====="); st1.collectPass(); st2.setLocation("Mirpur"); st1.showDetails(); st2.showDetails(); System.out.println("9====="); bus1.board(st1); bus1.board(st2, st3); System.out.println("10====="); bus1.showDetails(); } }</pre>	<pre>1===== Student Name: Afif Lives in Mirpur Have Bus Pass? false Student Name: Shanto Lives in Motijheel Have Bus Pass? false 2===== Student Name: Taskin Lives in Mirpur Have Bus Pass? false 3===== Bus Route: Mirpur Passenger Count: 0 (Max: 2) Passengers on Board: Bus Route: Azimpur Passenger Count: 0 (Max: 5) Passengers on Board: 4===== 5===== Student Name: Shanto Lives in Motijheel Have Bus Pass? true Student Name: Taskin Lives in Mirpur Have Bus Pass? true 6===== No passengers 7===== You don't have a bus pass! You got on the wrong bus! 8===== Student Name: Afif Lives in Mirpur Have Bus Pass? true Student Name: Shanto Lives in Mirpur Have Bus Pass? true 9===== Afif boarded the bus. Shanto boarded the bus. Bus is full! 10=====</pre>

	Bus Route: Mirpur Passenger Count: 2 (Max: 2) Passengers on Board: Afif Shanto
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Task 4

1	public class ExamClass {
2	public int ques;
3	public int sum;
4	public void methodB() {
5	System.out.println(ques + " " + 0 + " " + 0);
6	}
7	}
8	class QuizA {
9	public int x, y;
10	public int sum = 1;
11	public QuizA(int x, int y) {
12	this.x = y;
13	this.y = x;
14	}
15	public void methodA() {
16	int x = 3;
17	y = this.y + x;
18	ExamClass exam = new ExamClass();
19	exam.sum = x;
20	exam.ques = this.y;
21	x = this.x + x + exam.sum;
22	this.y = this.sum + methodB(exam.ques, exam);
23	System.out.println(x + " " + this.y + " " + sum);
24	sum = x % 2 + this.x;
25	y = x + y + exam.sum;
26	System.out.println(x + " " + y + " " + sum);
27	}
28	public int methodB(int x1, ExamClass x2) {
29	int y = 0;
30	y = this.y + x2.sum;
31	x2.ques = x1 + x2.ques;
32	sum = sum + x + y;
33	System.out.println(this.x + " " + this.y + " " + sum);
34	return x2.sum;
35	}
36	}

Driver Code	Output		
<pre> public class QuizTesterA{ public static void main(String []args){ QuizA q1 = new QuizA(3,4); q1.methodA(); } } </pre>			

HOMEWORK

Task 1

Design the Company and Employee classes so that the Tester1 class produces the given outputs.

Restriction: Company class can't have more than 1 array and all attributes of Employee class must be Private.

Driver Code	Output
<pre>public class Tester1{ public static void main(String args[]){ Employee e1 = new Employee(); Employee e2 = new Employee("Alif", 34, "Fulltime"); System.out.println("1-----"); Company c1 = new Company(); c1.details(); System.out.println("2-----"); Employee e3 = new Employee("Akter", 36,"Part-time"); Employee e4 = new Employee("Ria", 38, "Fulltime"); System.out.println("3-----"); c1.addEmployee(e2); c1.addEmployee(e3); System.out.println("4-----"); c1.details(); System.out.println("5-----"); c1.addEmployee(e4); c1.addEmployee(e1); System.out.println("6-----"); c1.details(); System.out.println("7-----"); c1.removeEmployee(e4); System.out.println("6-----"); c1.details(); } }</pre>	<pre>A default employee has been created 1----- Company Name: ABC Company Total Employee: 0 Fulltime Employees: Part-Time Employees: 2----- 3----- Alif has joined the company Akter has joined the company 4----- Company Name: ABC Company Total Employee: 2 Fulltime Employees: Name: Alif, ID: 34 Part-Time Employees: Name: Akter, ID: 36 5----- Ria has joined the company No more vacancy 6----- Company Name: ABC Company Total Employee: 3 Fulltime Employees: Name: Alif, ID: 34 Name: Ria, ID: 38 Part-Time Employees: Name: Akter, ID: 36 7----- Ria has left the company 6----- Company Name: ABC Company Total Employee: 2 Fulltime Employees: Name: Alif, ID: 34 Part-Time Employees: Name: Akter, ID: 36</pre>

Task 2

Please write the **Student** and **Department** class with the necessary properties so that the provided driver code generates the output given below. The id of the Student class will be **private**. For simplicity, assume that a department can add a maximum of 5 students.

Driver Code	Output
<pre>public class Tester1 { public static void main(String[] args) { Student s1 = new Student("Akib", 10, 3.29); Student s2 = new Student("Reza", 15, 3.45); Student s3 = new Student("Kabir", 20,4.0); System.out.println("1====="); Department cse = new Department("CSE"); cse.findStudent(-100); System.out.println("2====="); cse.addStudent(s1, s2, s3); System.out.println("3====="); cse.details(); System.out.println("4====="); cse.findStudent(15); System.out.println("5====="); Student s4 = new Student("Nakib", 15,3.22); cse.addStudent(s4); System.out.println("6====="); s4.setId(25); cse.addStudent(s4); System.out.println("7====="); cse.details(); System.out.println("8====="); Student s5 = new Student("Sakib", 30,2.29); cse.addStudent(s5); System.out.println("9====="); cse.details(); } }</pre>	<pre>1===== Student with this ID doesn't exist, Please give a valid ID 2===== Welcome to CSE department, Akib Welcome to CSE department, Reza Welcome to CSE department, Kabir 3===== Department Name: CSE Number of student:3 Details of the students: Student name: Akib, ID: 10, cgpa: 3.29 Student name: Reza, ID: 15, cgpa: 3.45 Student name: Kabir, ID: 20, cgpa: 4.0 4===== Student info: Student Name: Reza ID: 15 CGPA: 3.45 5===== Student with the same ID already exists, Please try with another ID 6===== Welcome to CSE department, Nakib 7===== Department Name: CSE Number of student:4 Details of the students: Student name: Akib, ID: 10, cgpa: 3.29 Student name: Reza, ID: 15, cgpa: 3.45 Student name: Kabir, ID: 20, cgpa: 4.0 Student name: Nakib, ID: 25, cgpa: 3.22 8===== Welcome to CSE department, Sakib 9===== Department Name: CSE Number of student:5 Details of the students: Student name: Akib, ID: 10, cgpa: 3.29 Student name: Reza, ID: 15, cgpa: 3.45 Student name: Kabir, ID: 20, cgpa: 4.0 Student name: Nakib, ID: 25, cgpa: 3.22 Student name: Sakib, ID: 30, cgpa: 2.29</pre>

Task 3

Spaceship: This class represents a spaceship. Each spaceship has a **name** and a **capacity** (the maximum weight it can carry).

Cargo: This class represents a piece of cargo. Each cargo item has a **name** and a **weight**. Both attributes should be **private** which means they cannot be accessed directly from outside of the class.

A **Spaceship** contains **Cargo**. That means each spaceship can carry multiple cargo items, but the total weight of the cargo cannot exceed the spaceship's capacity. Also, the maximum number of cargo items is 100. Your task is to design the **Spaceship** and **Cargo** class with necessary properties so that the given output is produced for the provided driver code.

Driver Code	Output
<pre>public class ShipTester { public static void main(String[] args) { Spaceship falcon = new Spaceship("Falcon", 50000); Spaceship apollo = new Spaceship("Apollo", 100000); Spaceship enterprise = new Spaceship("Enterprise",220000); System.out.println("1.====="); Cargo gold = new Cargo("Gold", 20000); Cargo platinum = new Cargo("Platinum", 25000); Cargo dilithium = new Cargo("Dilithium", 50000); Cargo trilithium = new Cargo("Trilithium", 70000); Cargo neutronium = new Cargo("Neutronium", 80000); System.out.println("2.====="); falcon.loadCargo(gold); falcon.loadCargo(platinum); falcon.displayDetails(); System.out.println("3.====="); apollo.loadCargo(gold); apollo.displayDetails(); System.out.println("4.====="); falcon.loadCargo(neutronium); System.out.println("5.====="); enterprise.loadCargo(dilithium); enterprise.loadCargo(trilithium); enterprise.loadCargo(neutronium); enterprise.displayDetails(); } }</pre>	<pre>1.===== 2.===== Spaceship Name: Falcon Capacity: 50000 Current Cargo Weight: 45000 Cargo:Gold Platinum 3.===== Spaceship Name: Apollo Capacity: 100000 Current Cargo Weight: 20000 Cargo:Gold 4.===== Warning: Unable to load Neutronium inside Falcon. Exceeds capacity by 75000. 5.===== Spaceship Name: Enterprise Capacity: 220000 Current Cargo Weight: 200000 Cargo:Dilithium Trilithium Neutronium</pre>

Task 4

Design the **necessary** classes for the UberEats system with the necessary properties to produce the given output for the provided driver code.

[All attribute of UberEatsUser class will be **Private**]

Driver code	Output
<pre>public class UberEatsAccount{ public static void main(String[] args){ System.out.println("====="); UberEatsUser user1 = new UberEatsUser("Peter Parker", "Badda"); UberEatsUser user2 = new UberEatsUser("Matt Murdock", "Mohammadpur"); UberEatsUser user3 = new UberEatsUser("Reed Richards"); UberEatsUser user4 = new UberEatsUser("Peggy Carter", "Mirpur"); Restaurant r1 = new Restaurant("Chillox", "Badda"); r1.takeOrder(user1); r1.takeOrder(user2); r1.takeOrder(user3); r1.takeOrder(user4); r1.completeOrders(); System.out.println("====="); Restaurant r2 = new Restaurant("Kyoshi", "Gulshan"); r2.takeOrder(user3); user3.updateLocation("Malibagh"); user3.updateLocation("Bashundhara"); r2.takeOrder(user3); r2.takeOrder(user3); r2.completeOrders(); r2.completeOrders(); System.out.println("====="); Restaurant r3 = new Restaurant("Cilantro","Banani"); r3.takeOrder(user1); r3.takeOrder(user2); r3.takeOrder(user3); r3.takeOrder(user4); } }</pre>	<pre>===== Your order has been added! Your order has been added! Location : Unknown. Please update your location information! Your order has been added! Showing Chillox's orders: Order by Peter Parker at Badda completed Order by Matt Murdock at Mohammadpur completed Order by Peggy Carter at Mirpur completed ===== Location : Unknown. Please update your location information! Update Successful! We already have your location. Please place an order! Your order has been added! You already have a pending order! Showing Kyoshi's orders: Order by Reed Richards at Malibagh completed No pending orders at the moment. ===== Your order has been added! Your order has been added! Your order has been added! We are really busy right now. Please order after some time. Thank you!</pre>

Task 5

1	public class msgClass{
2	public int content;
3	}
4	class FinalT5A{
5	public int sum = 2, y = 1, x = 1;
6	public void methodA(){
7	int x=6, y =0;
8	msgClass myMsg = new msgClass();
9	myMsg.content = this.x;
10	x = x + myMsg.content;
11	this.y = this.y + methodB(myMsg, myMsg.content);
12	System.out.println(x + " " + this.y+ " " + sum);
13	y = this.y/2 + this.x;
14	x = y + sum/2;
15	sum = x + y + myMsg.content;
16	System.out.println(x + " " + y+ " " + sum);
17	}
18	public int methodB(msgClass mg2, int mg1){
19	int x = 0;
20	y = y + mg2.content;
21	mg2.content = y + mg1;
22	x = this.x + 3 + mg1;
23	sum = sum + x + y;
24	System.out.println(this.x + " " + this.y+ " " + sum);
25	mg2.content = sum - mg1 ;
26	return sum;
27	}
28	}

DRIVER CODE	OUTPUTS		
<pre> public class Tester10{ public static void main(String args []){ FinalT5A fT5A = new FinalT5A(); fT5A.methodA(); } } </pre>			

Task 6

1	public class Foo{
2	public int bar, buz;
3	public Foo(int bar, int buz){
4	this.bar = bar;
5	this.buz = buz;
6	}
7	}
8	class Quiz5{
9	public int sum = 12, x = 2, y = 6;
10	public Foo foo;
11	public Quiz5(Foo f){
12	foo = f;
13	int x = this.foo.buz + y;
14	sum = sum + (f.bar--) + y;
15	System.out.println(foo.bar + " " + sum + " " + x);
16	sum -= 10;
17	}
18	public void methodA(int bar, int buz){
19	bar = 3 + bar - this.foo.bar;
20	x = bar + 12 + y;
21	y = foo.buz + buz + bar;
22	sum = y + methodB(foo.buz, foo) + foo.buz;
23	System.out.println(bar + " " + y + " " + sum);
24	}
25	public int methodB(int bar, Foo buz){
26	int sum = bar + buz.bar + x;
27	buz.buz = sum + this.sum;
28	System.out.println(bar + " " + buz.buz + " " + sum);
29	return sum;
30	}
31	}

Driver Code	Output		
<pre> public class LabTester{ public static void main(String []args){ Foo p = new Foo(13, 4); Quiz5 q = new Quiz5(p); q.methodA(14, 8); } } </pre>			